DESIGN AND SIMULATION OF SMART WATER TANK OVERFLOW PROTECTION AND PUMP CONTROL SYSTEM USING COMBINATIONAL LOGIC

GROUP-7

Introduction

- Water tank overflow leads to wastage of water and electricity.
- Manual pump control requires continuous monitoring.
- An automatic digital system can control the pump according to the water level.
- The proposed system uses **water-level sensors and combinational logic**.
- The system automatically switches the pump ON/OFF to prevent overflow.

OBJECTIVES

- Detect the water level inside the tank.
- Automatically control the water pump.
- Prevent water tank overflow.
- Reduce water and electricity wastage.
- Design the system using basic digital logic gates.
- Verify the design through simulation.

System Components and Block Diagram

Components

- Water-level sensors
- NOT/logic gates
- Pump-control circuit
- Relay
- Water pump
- LED indicators
- Water tank

Block Diagram



The sensor detects the water level and provides a digital input to the logic circuit.

components required

Hard ware components

- Water-level sensors/electrodes
- Logic gates
- Transistor/relay driver
- Relay
- Water pump
- LED indicators
- DC power supply
- Connecting wires
- Water tank

Simulation components

- Digital switches/sensor inputs
- Logic gates
- LED indicators
- Relay/pump representation
- Oscilloscope/logic analyzer if required

Simulation And Result

Simulation Procedure

1. Create sensor inputs using digital switches.
2. Connect the NOT and AND gates.
3. Connect the output to an LED/relay representing the pump.
4. Change the sensor inputs.
5. Observe and verify the pump output.

Advantages And Applications

Advantages

- Prevents water overflow.
- Reduces water wastage.
- Saves electrical energy.
- Reduces manual intervention.
- Simple and low-cost circuit.
- Easy to simulate and implement.

Applications

- Residential water tanks
- Apartments
- Schools and colleges
- Hospitals
- Industrial water-storage systems
- Agricultural water tanks

Conclusion And Future Scope

Conclusion

- A smart water-tank pump-control system was designed using **combinational logic.**
- Water-level sensors provide digital inputs.
- AND and NOT gates generate the pump-control signal.
- The system automatically turns the pump OFF when the tank is full.
- Simulation verifies the correct operation of the proposed circuit.

Future Scope

- IoT-based water-level monitoring
- Mobile notifications
- LCD/OLED water-level display
- Dry-run protection
- Flow sensors
- Automatic fault detection
- Microcontroller/PLC integration

THANK YOU